

Definition of linear ODE

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An n -th order ODE is *linear* if it can be written in the form

$$a_n(x)y^{(n)}(x) + \cdots + a_2(x)y''(x) + a_1(x)y'(x) + a_0(x)y(x) = b(x), \quad (1)$$

for some functions a_i for $i \in \{1, 2, \dots, n\}$ and b .

If the ODE cannot be written in that form, it is *nonlinear*.

We wanted to check the linearity or nonlinearity of the following ODEs:

- $u'(t) = u(t)$
- $u''(t) = (u'(t))^2 + 1$
- $y'(x) = 4(y(x))^2 + x^2$

The ODE $u'(t) = u(t)$ is linear because it can be written in the form (1) by choosing $a_1(x) = 1$, $a_0(x) = -1$ and $b(x) = 0$.

The ODE $u''(t) = (u'(t))^2 + 1$ is nonlinear because it cannot be written in the form (1). The issue is that the derivative u' is *squared*.

The ODE $y'(x) = 4(y(x))^2 + x^2$ is nonlinear because it cannot be written in the form (1). The issue is now that the function y is *squared*.